TOPIC 03

Probability Distributions

What a distribution is, discrete and continuous types, and the named families you will meet again and again

WHAT YOU WILL LEARN

- Explain what a probability distribution is, and why it is not the same as its graph
- Read a probability mass function (PMF), probability density function (PDF) and cumulative distribution function (CDF)
- Use the uniform, Bernoulli, binomial and Poisson distributions
- Know where the normal, t, chi-square and F distributions will appear
- Choose a sensible distribution for a real situation

1. What is a distribution?

DEFINITION **Distribution**

A **probability distribution** is a function that shows the possible values of a variable and how often (how likely) each occurs. In statistics, when someone says 'distribution' they mean a *probability* distribution. Examples: uniform, binomial, Poisson, normal, Student's t.

A common mistake is to think the distribution **is** the graph. It is not. The distribution is the *rule* that assigns probabilities. The graph is only a convenient picture of that rule, and every distribution has a characteristic shape that statisticians like to plot.

Two facts hold for every distribution: no probability is negative, and **all probabilities add up to 1 (100%)**.

2. Discrete distributions

A variable is **discrete** if its possible values can be listed (0, 1, 2, ... or the six faces of a die). Its distribution is described by a **probability mass function (PMF)**: P(X = x) for each possible x.

Example 1: one fair die

Each face has probability 1/6 ≈ 0.17. Impossible results, such as a 7, have probability 0. Because every outcome is equally likely, this is called a **uniform distribution**.

Outcome x	1	2	3	4	5	6	Anything else
P(X = x)	1/6	1/6	1/6	1/6	1/6	1/6	0
Decimal	0.17	0.17	0.17	0.17	0.17	0.17	0

Example 2: two fair dice

Now roll two dice and look at the **sum**. There are 6 × 6 = 36 equally likely combinations. A sum of 1 is impossible (the minimum is 1 + 1 = 2), a sum of 2 happens only as (1,1), a sum of 3 as (1,2) or (2,1), and so on.

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1
P	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36
Decimal	0.028	0.056	0.083	0.111	0.139	0.167	0.139	0.111	0.083	0.056	0.028

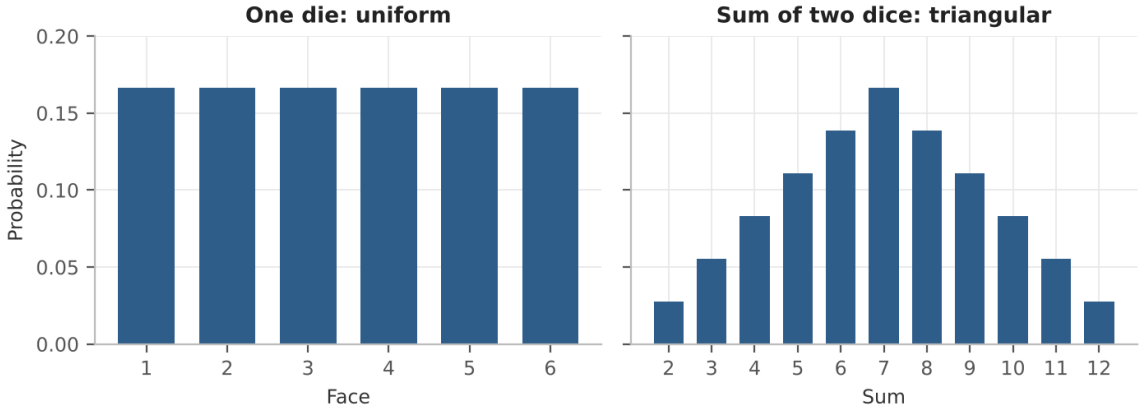


Figure 1. Same dice, different distributions. A sum of 7 can be made in six ways, so it is the most likely.

Notice what changed: the *experiment* is the same kind of thing, but the variable we measure differs, so the distribution differs. The distribution belongs to the **variable**.

3. Continuous distributions

A **continuous** variable can take any value in an interval (waiting time, height). There are infinitely many possible values, so the probability of hitting **one exact value** is 0. Instead we describe probability over *ranges* using a **probability density function (PDF)**, f(x).

$$P(a \le X \le b) = \int_a^b f(x)\,dx \qquad \text{(area under the curve)}, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

KEY IDEA **Area = probability**

For a continuous variable, **the height of the curve is not a probability; the area under it is.** The total area is 1. P(X = 5) = 0, but P(4.9 ≤ X ≤ 5.1) is a small positive area.

WORKED EXAMPLE **Bus waiting time (continuous uniform)**

A bus arrives at a random moment in a 10-minute window, so the waiting time X is uniform on [0, 10] with f(x) = 1/10. P(X ≤ 3) = 3 × (1/10) = 0.30 and P(4 ≤ X ≤ 7) = 0.30. The mean wait is 5 minutes.

The cumulative distribution function (CDF)

For any distribution the **CDF** is F(x) = P(X ≤ x), the probability of being at or below x. It rises from 0 to 1 and answers 'what fraction is below this value?'. It is also exactly what a z-table lists for the normal distribution.

	Discrete	Continuous
Possible values	Countable list	Any value in an interval
Described by	PMF: P(X = x)	PDF: f(x)
P(X = x)	Can be positive	Always 0
Probability of an event	Add the probabilities	Area under the curve (integral)
Examples	Dice, counts, yes/no	Time, height, weight, income

4. Important discrete distributions

Bernoulli: one yes/no trial

One trial with two outcomes: success (probability p) or failure (1 − p). Coded 1 and 0. Mean = p, variance = p(1 − p). Examples: one coin flip, one customer buys or not.

Binomial: count of successes in n trials

Count the successes in **n** independent Bernoulli trials with the same success probability **p**. Check the four conditions (BINS): **B**inary outcomes, **I**ndependent trials, fixed **N**umber of trials, same **S**uccess probability.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mu = np, \qquad \sigma^2 = np(1-p)$$

WORKED EXAMPLE **Ten coin flips**

X = number of heads in 10 fair flips, so X ~ Binomial(10, 0.5). P(X = 6) = C(10,6) $(0.5)^{10}$ = 210/1024 = **0.2051**. P(X ≤ 3) = **0.1719**. Mean = np = 5, SD = √(10 × 0.5 × 0.5) = 1.581.

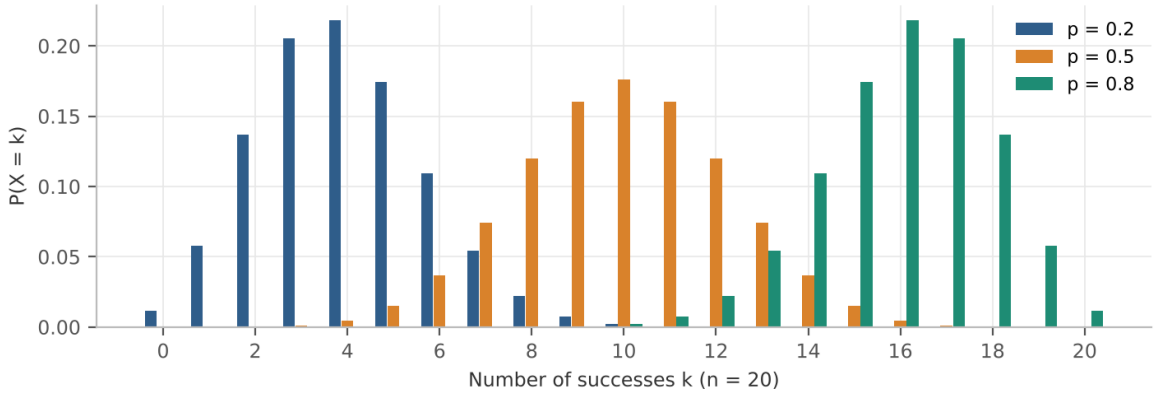


Figure 2. Binomial(20, p). Skewed right for small p, symmetric at p = 0.5, skewed left for large p.

Poisson: events in a fixed interval

Counts how many times a rare event occurs in a fixed time or space when events happen independently at an average rate **λ** (lambda). The mean and the variance are both λ.

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad \mu = \sigma^2 = \lambda$$

WORKED EXAMPLE **Support calls**

A help-desk receives on average 3 calls per hour. The probability of exactly 5 calls in an hour is e^{-3} 3^5 / 5! = **0.1008**. The probability of no calls is e^{-3} = **0.0498**. P(more than 5) = **0.0839**.

5. Important continuous distributions

Degrees of freedom (df) is a number that tells a distribution how much independent information the data held; it sets the exact shape (Topic 08 explains it). The t, chi-square and F distributions are used through tables or software, never by hand. When n is large the binomial is close to a normal curve; this normal approximation is covered in Topic 06.

Distribution	Shape and parameters	Typical use	Appears in this handbook
Uniform	Flat on [a, b]	Random numbers, waiting within a window	Topic 03, 06
Normal $N(\mu, \sigma^2)$	Symmetric bell curve	Heights, measurement errors, IQ, and *sample means* (Topic 06)	Topics 04 to 10
Student's t	Like normal but heavier tails; one parameter, degrees of freedom	Estimating a mean when σ is unknown, small samples	Topics 09, 10
Chi-square χ^2	Right-skewed, non-negative; degrees of freedom	Variances, goodness-of-fit, independence in tables	Topic 11
F	Right-skewed; two degrees of freedom	Comparing variances, ANOVA	Topic 11
Exponential	Right-skewed, starts high	Time between events (Poisson process)	Topic 06 (simulation)

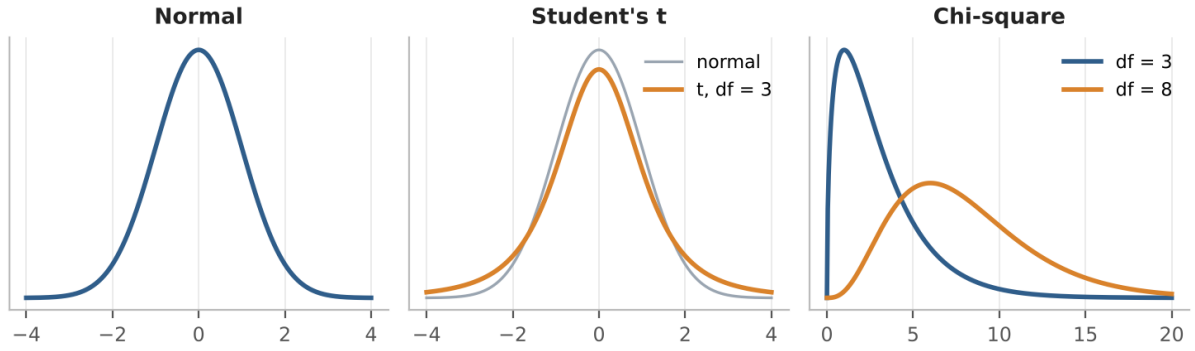


Figure 3. Three continuous families you will use. Each is defined by its density function, not by the picture.

6. Choosing a distribution

Situation	Distribution
One yes/no trial	Bernoulli
Number of successes in a fixed number of independent yes/no trials	Binomial
Number of events in a fixed time or area, at a steady average rate	Poisson
All values in a range equally likely	Uniform
Many small independent effects added up, or an average of many values	Normal
Mean of a small sample with unknown σ	Student's t
Waiting time between random events	Exponential

7. Summary table of means and variances

Distribution	Mean	Variance
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$
Poisson(λ)	λ	λ
Uniform(a, b)	$(a + b)/2$	$(b - a)^2/12$
Normal(μ, σ^2)	μ	σ^2

8. In Python

```
from scipy import stats

stats.binom(10, 0.5).pmf(6)        # P(X = 6)        -> 0.2051
stats.binom(10, 0.5).cdf(3)        # P(X <= 3)       -> 0.1719
stats.poisson(3).pmf(5)            # P(X = 5)        -> 0.1008
stats.uniform(0, 10).cdf(3)        # P(wait <= 3)    -> 0.3
stats.norm(0, 1).cdf(1.96)         # P(Z <= 1.96)    -> 0.9750
stats.binom(10, .5).rvs(5)         # five random draws
```

SUMMARY

- A distribution is the rule linking values to probabilities; the graph is just a picture.
- Discrete: PMF, probabilities add to 1. Continuous: PDF, area equals probability, $P(X = x) = 0$.
- Binomial counts successes in n independent trials; Poisson counts events at a rate λ.
- The normal, t, chi-square and F distributions are the workhorses of inference, and all are continuous.

Practice questions

Try each question before reading the answer underneath it.

Q1. Why is $P(X = 2.5) = 0$ for a continuous variable?

Answer. There are infinitely many possible values, so the probability of any one exact value is 0; only ranges have positive probability (area under the curve).

Q2. Is the sum of two dice uniformly distributed?

Answer. No. Sums have different numbers of ways (1 for 2, 6 for 7), so it is triangular, not uniform.

Q3. A fair coin is flipped 8 times. Find P(exactly 4 heads).

Answer. $C(8,4)/2^8 = 70/256 = 0.2734$.

Q4. A shop gets 2 customers per 10 minutes on average. P(none in 10 minutes)?

Answer. Poisson, $\lambda = 2$: $e^{-2} = 0.1353$.

Q5. A random variable is uniform on [0, 20]. What is $P(5 \le X \le 12)$?

Answer. $(12 - 5)/20 = 0.35$.

Q6. Which has the larger variance: Binomial(100, 0.5) or Binomial(100, 0.9)?

Answer. $np(1-p)$: 25 versus 9, so $p = 0.5$ has the larger variance.